

THE FINEST TECHNOLOGY STP EVER HAD....



Full and Full biological wastewater treatment plant using the SBR (Sequence Batch Reactor) process with upstream Collection Tank

TWO - STAGE SBR PROCESS

1. Stage: Raw Sewage Collection tank / Equalization tank

- Primary and Secondary sludge storage
- Retention of solids and floating matters
- Volume Balancing and concentration related fluctuations in the waste water inflow.

2. Stage: Activated sludge process operated in single reactor in batch

- Biological clarification with activated sludge
- Nitrification and denitrifications
- Phosphate precipitation (optional)

OPERATING MODE

- Real - time control via PLC with process flow settings
- Standard setting: 4 Purification Batches of 6 hours each per day
- Process flows Display Via HMI

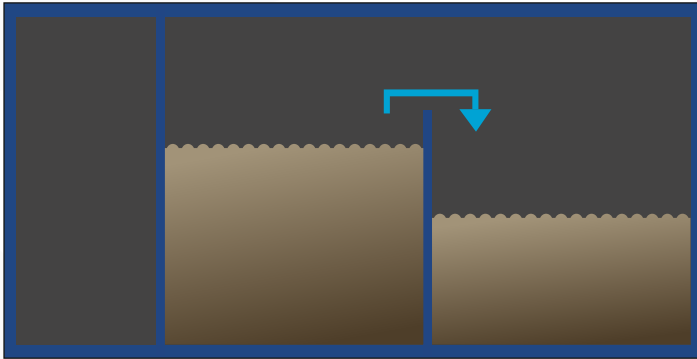
OPERATING MODE

- Waste water Aeration via Fine Bubble Diffuser
- Air Lift Technology For Filling, Clear Water Removal & Sludge Re-Cycle
- Smart Mode - Possible In Under load

ADVANTAGE

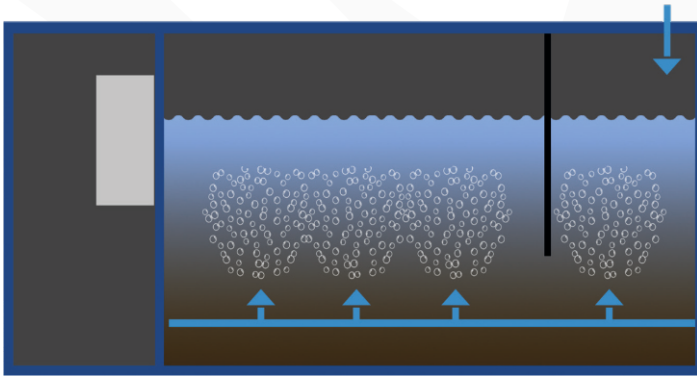
- Operator Not Required
- 35% Power Saving, compared to other technology
- Without pumps process functions
- Installation In CONCRETE, FRP & MSFRP TANKS
- Stable process as compared to hydraulic shock loads and underload
- High reliability and low maintenance
- User-friendly due to fully automatic plant





1. LOADING PHASE

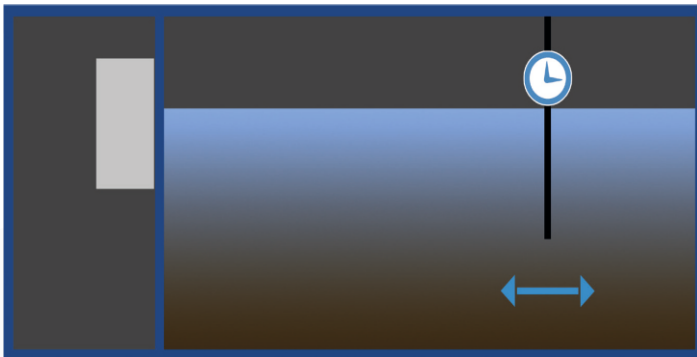
The raw sewage water stored in the Collection/Equalization tank is lifted to SBR through an air lift arrangement. The special design of the airlift ensures a required water level in the collection tank. Limiting the water level by incorporating other components (e.g. Pump & float switch) is not required.



2. AERATION PHASE

The sewage water is aerated and mixed with fine-bubble diffusers fixed at bottom of the tank. The compressed air required for aeration is produced by an air blower. Aeration generally takes place intermittently. Two effects are achieved simultaneously by this aeration.

- ▶ The micro-organisms in the activated sludge are supplied with oxygen, which is necessary for their metabolic activity and for the reduction of pollutants.
- ▶ It brings the bacteria in close contact with the wastewater.



3. SETTLE PHASE

No aeration takes place during the settling mode. The activated sludge is allowed to settle. A sludge blanket forms on the tank bottom and in the upper layer of a clear water zone



4. CLEAR WATER EXTRACTION AND SLUDGE RETURN

The clear water is transferred from the SBR stage using air lift technology.

The special design of this air lift technology

- ▶ Without additional Mechanical Equipment (Decanter) the clear water removed from SBR tank with airlift technology.
- ▶ Excess sludge also returned to the sludge storage tank through the air lift technology. (Additional sludge pump not required)

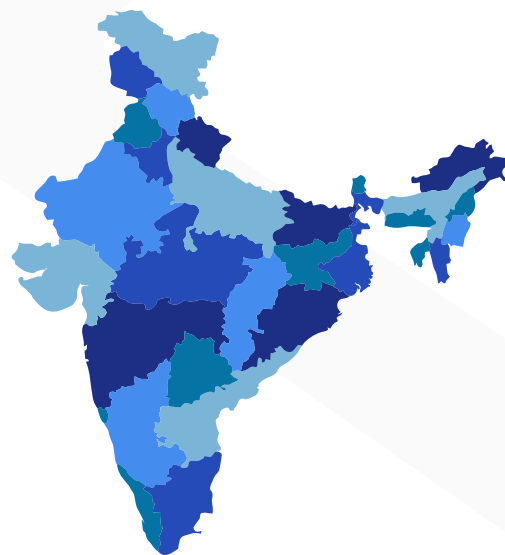


WASTEWATER PARAMETER	IKON EFFLUENT VALUES*	DEGREE OF EFFICIENCY
COD (chemical oxygen demand)	48 mg/l	92.3 %
BOD (biochemical oxygen demand)	6 mg/l	97.5 %
NH4-N (ammonium nitrate)**	8.3 mg/l	75.8 %
Nottot (total nitrogen)**	16 mg/l	67.5%
SS (suspended solids)	7 mg/l	96.7 %

IKON SMART SBR - SEWAGE TREATMENT PLANT

Starts from 1 KLD to 1 MLD

- ✓ **Current Indian market leader for waste water treatment plants**
- ✓ **Experience since 2011**
- ✓ **Small sized company in India**
- ✓ **IKON is a Group of Indian Water Technologies**
- ✓ **German design and engineering**
- ✓ **Wastewater treatment plants from 1 KLD to 1 MLD**
- ✓ **Solutions for wastewater reuse**
- ✓ **Grease- and light fluid separators**



More than 1000 plants all over INDIA

- Ikon Smart SBR - Monitors fill level
- In Absence or Low inlet a cycle will adjust

- System is marginally aerated to maintain bacteria
- Saves Energy



FLEXIBLE

- ▶ Easy to transport
- ▶ Prefabricated and expandable design
- ▶ Can be connected in parallel to process larger volumes
- ▶ Stackable and transportable
- ▶ Flexibly expandable (e.g. Ikon Smart Web Monitor, disinfection...)
- ▶ Suitable for both long-term and temporary use

EFFICIENCY

- ▶ Low energy consumption
- ▶ Fully automated and user-friendly
- ▶ Wear-resistant components for low maintenance
- ▶ Quick to install and remove (plug-and-play)

**Add-on modules**

Guardrail with impact protection

Phosphate precipitation

Chlorine disinfection

UV disinfection

Sludge dewatering



APPLICATIONS

Ikon Pre-Fab was developed as a temporary solution but can also be used as a permanent sewage treatment plant.

- ▶ Labourer Camps
- ▶ Temple, Festival & Events
- ▶ Factory water treatment at industrial park
- ▶ Seasonal tourism
- ▶ Construction sites for mobile use

DESIGN CRITERIA

The Ikon Pre-Fab plant is designed based on Indian regulations and standards for wastewater treatment. The design factors in both hydraulic and organic loads as well as the required treatment efficiency. The components are designed based on these requirements.

RAW WASTEWATER

This plant is designed for the following wastewater values:

Total suspended solids TSS	150 – 400 mg/l
BOD ₅	150 – 500 mg/l
COD	300 – 1,000 mg/l
pH	7.5 - 8.5
N _{tot}	20 – 80 mg/l
P _{tot}	6 – 25 mg/l

CLARIFIED WASTEWATER

The quality of the treated wastewater is normally within or below the following ranges:

pH	7.0 - 8.0
SS	< 30 mg/l
BOD ₅	< 10 mg/l
COD	< 120 mg/l
NH ₄ -N	< 10 mg/l
P _{tot}	< 20 mg/l
E. coli	< 100 cfu
Enterococcus	< 100 cfu



Raw Sewage Water



Treated Water



Controllers

Ikon Smart PLC offers proprietary, sophisticated controllers, which are widely popular in the branch of industry. The devices have been constantly further developed and improved by us over the years, though the structure and the concept has been retained. The controllers are easily understandable, very user-friendly.



HARDWARE

- ▶ Large Graphic HMI Display With Background Lighting
- ▶ Process Status - Graphical FLOW CHART
- ▶ Readout Of Operating Data
- ▶ Visual And Acoustic Alarm
- ▶ Voltage Monitoring Relay
- ▶ Energy Meter
- ▶ Multifunction Meter – Voltage, Current & Frequency
- ▶ Extern. Alarm
- ▶ Output Rs 232 For Communication
- ▶ Temperature Sensor And Pressure Sensor
- ▶ All Connections Plug-in Type

SOFTWARE

- ▶ Real-time Control
- ▶ With Preset, Editable Work Cycles
- ▶ Monitoring Functions (power Failure, Compressor, Valves, Temperature, Pressure, Etc.)
- ▶ Comprehensive Logbook Function
- ▶ Ease Of Operation
- ▶ Operating hours meter and manual operation function, economy mode, logbook view, etc.
- ▶ Service layer (code-protected):
- ▶ Cycle times and operating sequences, dosing and Auto Backwash for Post treatment Filters restart and function test, monitoring functions ON/OFF, Service, etc.





350 KLD SEWAGE TREATMENT PLANT



AUTOMATIC DUAL MEDIA FILTER



150 KLD SEWAGE TREATMENT PLANT



100 KLD SEWAGE TREATMENT PLANT



40 KLD SEWAGE TREATMENT PLANT

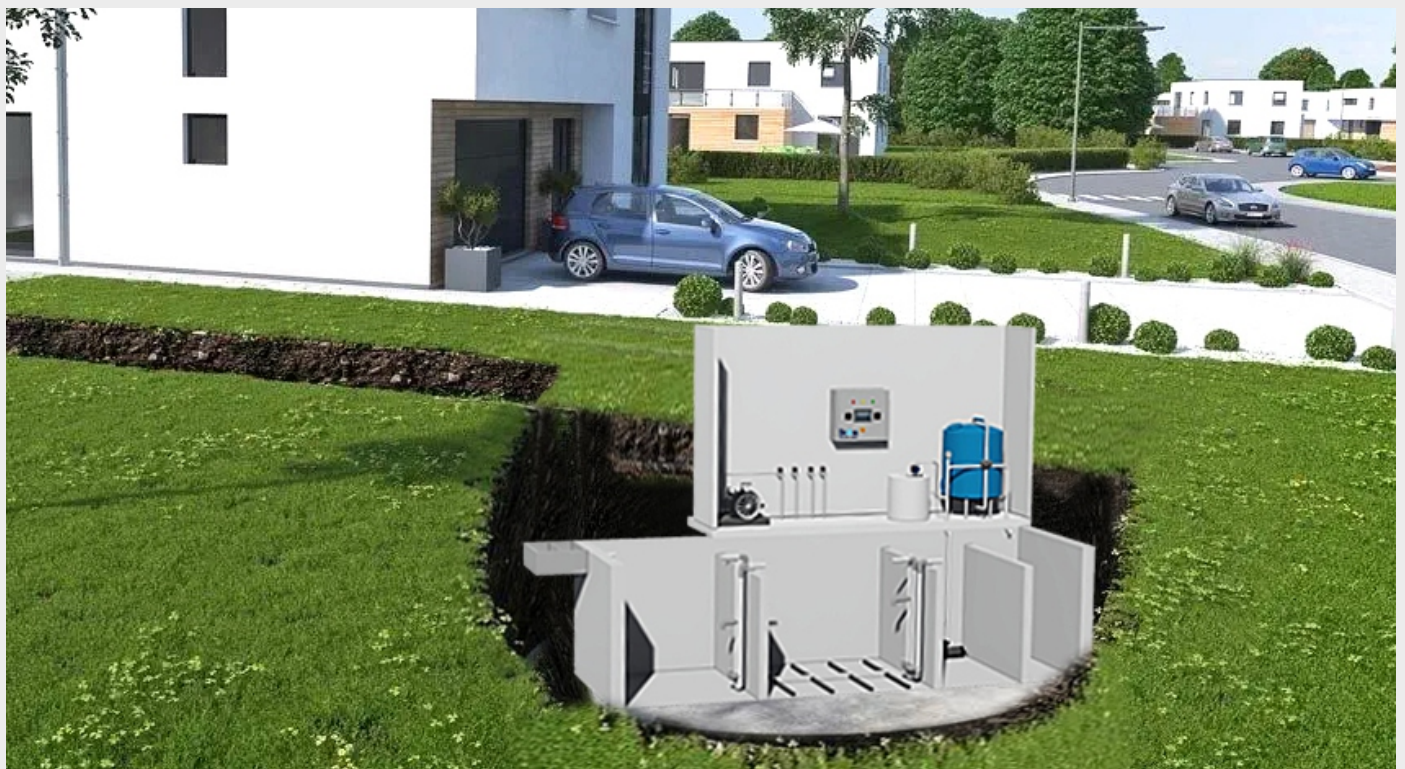


350 KLD SEWAGE TREATMENT PLANT



ALVAS MEDICAL COLLEGE Mudipu, Karnataka	MITE ENGINEERING COLLEGE Mudipu, Karnataka	MANGALORE UNIVERSITY Mangalore, Karnataka
CITADEL DEVELOPERS Mangalore, Karnataka	LANDMARK Mangalore, Karnataka	SURYA INFRA TECH Mangalore, Karnataka
KATEEL TEMPLE Kateel, Karnataka	KURLON LTD Bangalore, Karnataka	TATA CONSUMABLES LTD Bangalore, Karnataka
PREETHI DEVELOPERS Coimbatore, Tamil Nadu	ANNAI VELANKANNI HOSPITAL Thirunelveli, Tamil Nadu	STAR PALACE HOTEL Rameshwaram, Tamil Nadu
HOTEL TAMARA Kumbakonam, Tamil Nadu	ACE ELECTRICALS Hosur, Tamil Nadu	HPCL Soolagiri, Tamil Nadu
SUPRAJITH ENGG LTD Bangalore, Karnataka	BOSCH REXROTH LTD Bangalore, Karnataka	GREEN APPLE APARTMENT Bangalore, Karnataka
AVON PLASTICS PVT LTD Bangalore, Karnataka	SAKTHI PRECISION PVT LTD Bangalore, Karnataka	VATIKA PRIDE APARTMENT Bangalore, Karnataka
AB CARTER LTD Bangalore, Karnataka	BIG BAGS INTERNATIONAL LTD Bangalore, Karnataka	HEBRON PROPERTIES Bangalore, Karnataka
RAJDHANI INDUSTRIAL PARK Bangalore, Karnataka	EVERSHINE GREENWOOD APARTMENT Bangalore, Karnataka	SHRIRAM DEVELOPERS Bangalore, Karnataka
TRIDENT GROUP Bangalore, Karnataka	MOUNT CARMEL COLLEGE Bangalore, Karnataka	GARDENCITY UNIVERSITY Bangalore, Karnataka





Advantages and features of the IKON SMART SBR System

Robust

- No risk of concrete corrosion
- Separation of waste water and technology
- Air pump instead of submersible pump
- Retrofit kit made of waste water resistant plastic / installation parts made of waste water resistant stainless steel

Safe

- No risk of decay odor
- Stable procedure against shock loads and under load
- Stable cleaning performance even in winter
- High operational reliability
- Mature technology - on the market since 2015 - in use worldwide

Flexible

- No tight partition wall required
- Independent of container material and geometry
- Wide range of process customization options via the control system

Efficient

- Sludge stabilization and mineralization, no digestion
- Very low sludge accumulation and thus low desludging frequency
- Low maintenance - no sludge measurement in primary clarification necessary
- Low energy consumption through fine bubble ventilation
- User-friendly, because of automatic operation



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GROUP OF COMPANIES

- ▶ Indian Water Technologies
- ▶ Suki management and consultancy services

BRANCHES

- ▶ Chennai
- ▶ Mangalore